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Working title: Effects of Comprehension-Paced Collaborative Problem-Solving on Performance and Self-Efficacy in a Numerical Methods Course for Chemical Engineering Students

Target journal: Education for Chemical Engineers

1. Introduction

Computational competence is strictly needed in modern chemical engineering curricula, owing to the discipline's broader shift toward simulation- and data-driven practice — a shift alumni, digitalisation experts, and academic staff converge on almost unanimously across three linked institutional surveys, with digital-model competency expected at a level above that asked of statistical or machine-learning content (Udugama et al., 2022), consistent with curriculum-level practice already embedding simulation software directly into core technical content delivery (e.g., Johnson and Singh, 2018), and with Excel-based tools and scripting languages such as Python rated by IChemE-accredited institutions as the best-suited platforms for teaching core quantitative subjects precisely because of their low mathematical and programming overhead (Udugama et al., 2023). However, for students who are not majoring in computer science, a numerical methods course rarely tests that competence alone. The numerical or algorithmic concept — for instance, the convergence behavior of an iterative root-finding method — is to be learned simultaneously with the coding syntax needed to implement it, whether in Excel or in Python. This simultaneity constitutes a compound cognitive load in a specific, well-documented sense: a novice programmer must build a working mental model of how the code will execute, learn the target language's notation, assemble reusable solution structures, and manage the practical mechanics of writing, testing, and debugging, several of which are largely independent of whatever numerical concept the code is meant to express (Robins, 2019). Held together with the numerical concept itself, these demands raise the number of interacting elements a novice must track at once substantially beyond what either demand would impose alone; thus, for students entering with low programming self-efficacy, the syntax burden may crowd out the conceptual learning it is meant to support. Note that numerical methods occupies an unusual position in the curriculum in this respect. Differently from a course such as process control, in which software functions merely as a delivery vehicle for content that could, in principle, be taught without it, numerical methods renders the coding burden and the technical-content burden simultaneous and inseparable. It is this compound load, rather than topical convenience, which motivates the course context chosen here.

A pedagogical format responding directly to this compound load should, in principle, separate the two burdens rather than requiring every student to carry both at once. Self-efficacy theory (Bandura, 1977) offers a named account of why such a separation may matter: perceived personal efficacy is built from four information sources, i.e., mastery experience (direct performance accomplishment), vicarious experience (observing a similar other succeed), verbal persuasion, and physiological or emotional state, and the more direct the source, the larger its effect. Consider a classroom format in which a single student operator carries the coding syntax whereas the rest of the class collectively directs the problem-solving, with the instructor's tempo responsive to whether conceptual understanding is landing. Such a format plausibly supplies two of these four sources within a single classroom event: the operator receives a direct mastery experience, while classmates observing a same-status peer succeed under low-stakes, collectively supervised conditions receive vicarious experience. This mechanism is the theoretical anchor for the self-efficacy outcome examined below. Moreover, it is consistent with the one available domain-specific test of Bandura's model in engineering populations, which finds mastery experience the most frequently cited source of engineering self-efficacy. Nevertheless, visible performance moments possess their own handicaps, being nearly as likely to lower self-efficacy as to raise it, which a design making problem-solving visible to the whole class must anticipate rather than assume away (Chen et al., 2023).

Two literatures already pace instruction by collective comprehension; yet, neither isolates the condition just described. Peer Instruction structures a lecture as short presentation segments, each followed by a conceptual multiple-choice question which students answer individually, discuss with neighbors, and answer again prior to the instructor's resolution (Mazur, 1997; Crouch and Mazur, 2001). Meta-analytic evidence across STEM disciplines confirms a positive average effect on learning gain, albeit one varying sharply by discipline and by national context (Balta et al., 2017), while a review restricted to health-professions education found large within-session conceptual gains without a corresponding advantage over other active methods on downstream exam performance (Aldalou et al., 2026). Process-Oriented Guided Inquiry Learning (POGIL) instead organizes self-managed small groups around a three-phase learning cycle, i.e., exploration, concept invention, and application, with the instructor acting as facilitator rather than as source of information (Moog and Spencer, 2008). Across 21 studies it shows merely a small average effect on continuous achievement, yet nearly doubles the odds that a student passes the course outright (Walker and Warfa, 2017), a pattern reproduced by a non-randomized trial in an undergraduate science course employing the same pretest-covariate strategy adopted here (Udu et al., 2020). Note that the two mechanisms differ in where comprehension is read: in Peer Instruction it is sampled through discrete checkpoint cycles distributed across individual pairs, whereas in POGIL pacing is set by structured materials progressing largely independently across small groups. Neither reproduces a single, whole-class, live problem-solving session directed collectively at one shared operator's workstation, with tempo set continuously by the instructor's real-time reading of the class as a whole.

A second pair of literatures raises pacing along a different axis, i.e., self-direction, and here the evidence is not straightforwardly consistent. In flipped classrooms, solely the pre-class content is self-paced by nature, whereas in-class time itself remains instructor-paced; this is the dominant model underlying most so-called self-paced language in the chemical engineering education literature, including a recent hybrid-flipped numerical methods course in which engagement level, not pacing structure, moderated outcomes, since high-engagement students showed significantly improved mastery confidence while low-engagement students showed none (Lewin and Barzilai, 2023). Fully asynchronous online formats remove instructor supervision altogether. What evidence exists suggests that it is the pacing behavior itself, rather than merely the absence of supervision, which matters: the regularity with which undergraduates space their work — not how quickly they start or finish — is the strongest behavioral predictor of both final grade and withdrawal (Lim, 2016), and a fully deadline-free policy produces the highest incompletion rate and the most end-of-term work clustering of any pacing policy tested in a large multi-term dataset (Chiu et al., 2024). Since that regularity is set exclusively by the individual learner, self-regulation is left to do work which a group-paced, instructor-observed format does not ask of any single student, a difference mattering notably more for students with weaker self-regulation than for stronger ones (Pogorskiy and Beckmann, 2023). This sits awkwardly against mastery-learning evidence: a 108-study meta-analysis found that group-paced, teacher- or system-responsive mastery programs produced significantly larger achievement gains than self-paced ones, and that self-pacing specifically, rather than mastery learning in general, was the variable associated with reduced course completion (Kulik et al., 1990). Read together, the two literatures suggest that the reconciling variable is not supervision alone but the locus of pacing control, i.e., whether pacing responds to the group's collective understanding or is left to each student's individual self-regulation. Note that this locus matters unequally across students, since self-regulated learning is not a single undifferentiated capacity: the Motivated Strategies for Learning Questionnaire separates the motivational beliefs a student brings to a course from the learning strategies deployed within it, scoring metacognitive self-regulation, effort regulation, and peer learning as distinct subscales rather than as facets of one trait (Pintrich et al., 1991, 1993), and these components shift differentially across a term (Esnaashari et al., 2025) and discriminate substantially between instructional conditions in undergraduate engineering (Cepeda-Argüelles et al., 2026). A format supplying external pacing structure hence does not act on self-regulation as such, but plausibly substitutes for some components while leaving others untouched; thus, self-regulated learning is treated here as a measured moderator rather than as an assumed constant.

None of these four adjacent literatures — mastery learning, flipped classroom, fully asynchronous self-paced online learning, and Peer Instruction/POGIL — isolates the classroom condition examined here: whole-class, single-student-operator, instructor-observed, comprehension-paced collaborative problem-solving, in which the manipulated variable is classroom structure itself rather than any particular comprehension-checking technique. Table 1 states that contrast property by property against each of the four. Note that the closest methodological neighbors fall short of this condition in identifiable ways. A sequential live-coding format has multiple selected students take turns coding while the class watches and the instructor backward-lectures afterward, which is turn-taking among selected individuals rather than continuous whole-class direction of one shared operator (Cho et al., 2024). Similarly, student-led tutorials have been employed explicitly to help students keep pace with a challenging course, albeit through peer-led small tutorials rather than a single-operator, whole-class format (Serra et al., 2023), whereas cooperative problem-based learning applies small groups of three to five students with a floating facilitator in a chemical engineering process control and dynamics course, which is small-group facilitation rather than whole-class direction of one shared artifact (Mohd-Yusof et al., 2011). Closest in discipline, a Peer Instruction trial on a quantitative chemical engineering course structurally similar with numerical methods obtained significant exam-score and engagement gains over traditional lecture, yet through the paired vote-discuss-revote structure (Teixeira et al., 2018). Closest in venue, a motivational active-learning design published in this journal combines Peer Instruction, flipped classroom elements, and gamification for process control instruction, still relying on discrete clicker-based Peer Instruction rather than on a live, single-operator, whole-class format (Rodríguez et al., 2018/2019).

Table 1. The comprehension-paced collaborative format contrasted with the four adjacent instructional literatures, on the four properties that jointly constitute it. No adjacent format combines all four; the rightmost three columns each differ on at least two.

Property	This study's format	Peer Instruction	POGIL	Flipped classroom	Asynchronous self-paced online
Session organization	Synchronous, whole class acting as one unit	Synchronous, whole class segmented into pairs	Synchronous, self-managed small groups (3–5)	Synchronous in class; asynchronous pre-class	Asynchronous, individual
Solution artifact	One shared artifact, projected, under construction by the class	One answer per student, then per pair	One worksheet per small group	One per student during in-class problem-solving	One per student
How comprehension is read	Continuously, informally,	Sampled at discrete	Embedded in the designed	Not systematically	Not read; no observer present

Property	This study's format	Peer Instruction	POGIL	Flipped classroom	Asynchronous self-paced online
	across the whole class	ConcepTest checkpoints	materials and group roles	read during class time	
Locus of pacing authority	Instructor, reading the collective	Instructor, at checkpoint intervals	The materials and each group's own progress	Instructor (in class); learner (pre-class content only)	The individual learner
Supervision present while work is done	Yes, continuous	Yes	Yes, as facilitator	Yes (in class)	No
Canonical sources	— (this study)	Mazur (1997); Crouch and Mazur (2001)	Moog and Spencer (2008)	Lewin and Barzilai (2023)	Lim (2016); Chiu et al. (2024)

Herein, we examine whether comprehension-paced collaborative problem-solving, compared with traditional instructor-paced instruction, improves exam performance (H1) and self-efficacy (H2) of chemical engineering students in a numerical methods course, across five sections of a single academic term ($N \approx 200$). This specific classroom format has not been tested in a multi-section, quasi-experimental design to the best of our knowledge. Self-regulated-learning moderation (H3), topic-level differential benefit (H4), and an optional motivation and self-efficacy trajectory analysis (H5) are examined as secondary, exploratory extensions rather than as claims a single-term sample is asked to carry on its own. The resulted evidence is intended to separate the effect of pacing locus from that of any particular comprehension-checking technique, which is of practical advantage for instructors choosing among active-learning formats under ordinary course constraints.

Please review this section for unintended similarity to source texts before proceeding.

2. Materials and methods

Draft status note (not for submission): the study described below has not yet been executed. Every procedural statement is the *planned* protocol as fixed in the Experiment Blueprint and the four instrument files, written in the past tense the final manuscript will carry, and must be verified against what was actually implemented before submission. Items still genuinely open are marked with an inline flag rather than written as though settled.

2.1. Participants and course context

The study was conducted across five sections of an undergraduate Numerical Methods course for chemical engineering students within a single academic term, comprising approximately 40 students per section ($N \approx 200$ in total). All five sections shared one instructor (the first author), one syllabus, one topic sequence, and one set of summative assessments; thus, instructor identity, content coverage, and assessment difficulty were held constant across conditions by design rather than adjusted statistically. Six instructional sessions were covered in the period studied, each pairing a numerical-methods topic with a matched spreadsheet (Excel) and Python implementation: linear equation systems by Gauss–Seidel and Jacobi iteration; linear systems applied to one-dimensional steady-state conduction; analytic treatment of nonlinear equations by Horner's method and polynomial root extraction; numerical root finding by bisection, secant, and Newton–Raphson; nonlinear systems solved for multi-reaction extent of reaction; and regression for kinetic parameter estimation, respectively.

Three sections were assigned to the comprehension-paced collaborative problem-solving condition and two to the traditional instructor-paced control condition. Assignment was made at the section level rather than at the individual level; hence, the design is quasi-experimental rather than randomized. Two features of section formation bear on the interpretability of that assignment and are stated explicitly here rather than left to inference. First, the five section rosters were formed through the institution's ordinary registration process, i.e., student choice constrained by timetable and system allocation, and not by any researcher-controlled selection, so that no roster was assembled with reference to prior achievement. Second, condition was assigned to whole sections by an arbitrary rule fixed in advance (section number and time-of-day rotation) which is unrelated to student ability, prior performance, or any characteristic plausibly correlated with the outcomes. Neither feature renders the design randomized. Baseline equivalence across sections is therefore treated as a claim to be tested rather than assumed, and the pretest battery described in Section 2.6 exists in part to support that test. Fig. 1 depicts the design, condition assignment, and measurement schedule as a whole.

Numerical Methods course • one term • one instructor • 5 sections • $N \approx 200$
 |
 rosters formed by ordinary student registration

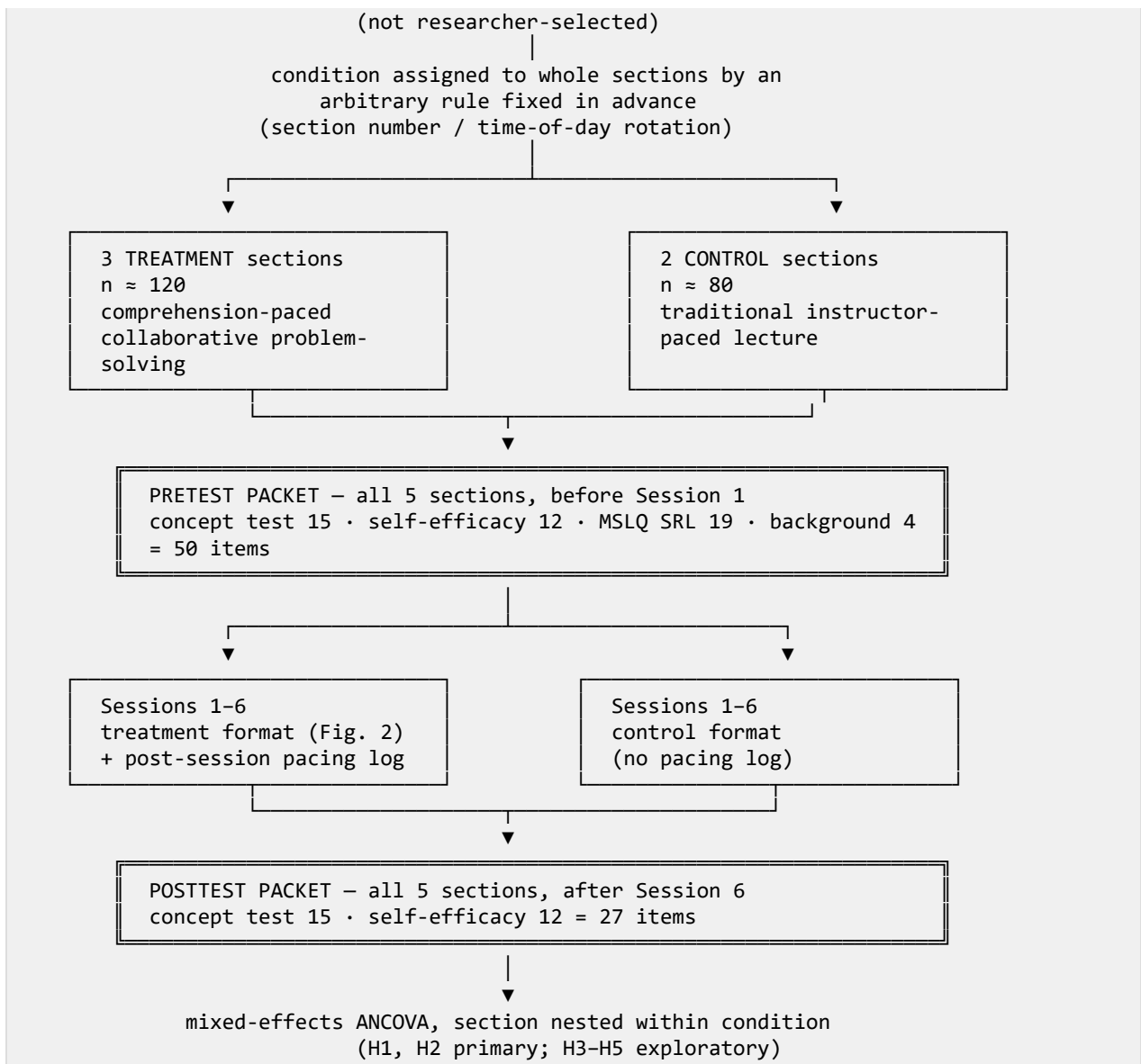
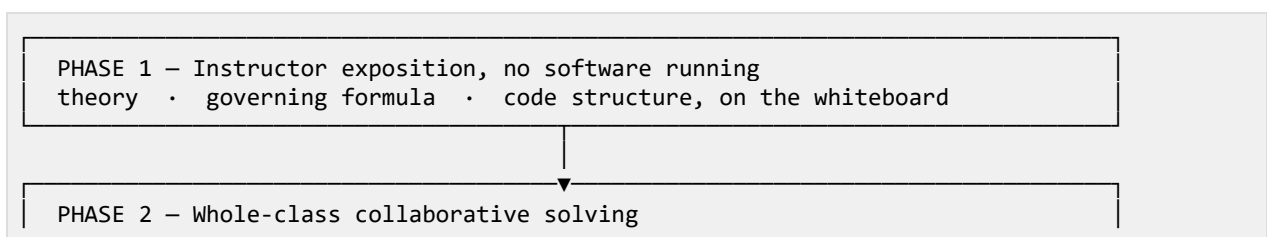


Fig. 1. Study design, condition assignment, and measurement schedule. Both conditions receive the identical syllabus, topic sequence, instructor, and summative assessments; the pretest and posttest packets are administered identically across all five sections. [ASCII placeholder — to be redrawn as a publication figure. Participant flow and attrition numbers to be added once data collection is complete, following CONSORT-style reporting.]

2.2. Operational definition of the intervention

The manipulated variable in this study is classroom *structure*, not a comprehension-checking technique. The treatment condition is defined as whole-class, single-shared-operator, instructor-observed, comprehension-paced collaborative problem-solving: one student at a time operates the software at a single workstation projected to the room, the class as a whole directs that operator's steps, and the tempo at which the problem advances is set by the instructor's continuous reading of whether the class collectively understands the current step. The four properties constituting this definition — synchronous whole-class organization, a single shared artifact, continuous rather than checkpoint-sampled comprehension reading, and instructor-held pacing authority — are the same four contrasted against the adjacent literatures in Table 1; Fig. 2 shows how they interact within a single session via the pacing feedback loop that is the format's defining mechanism.



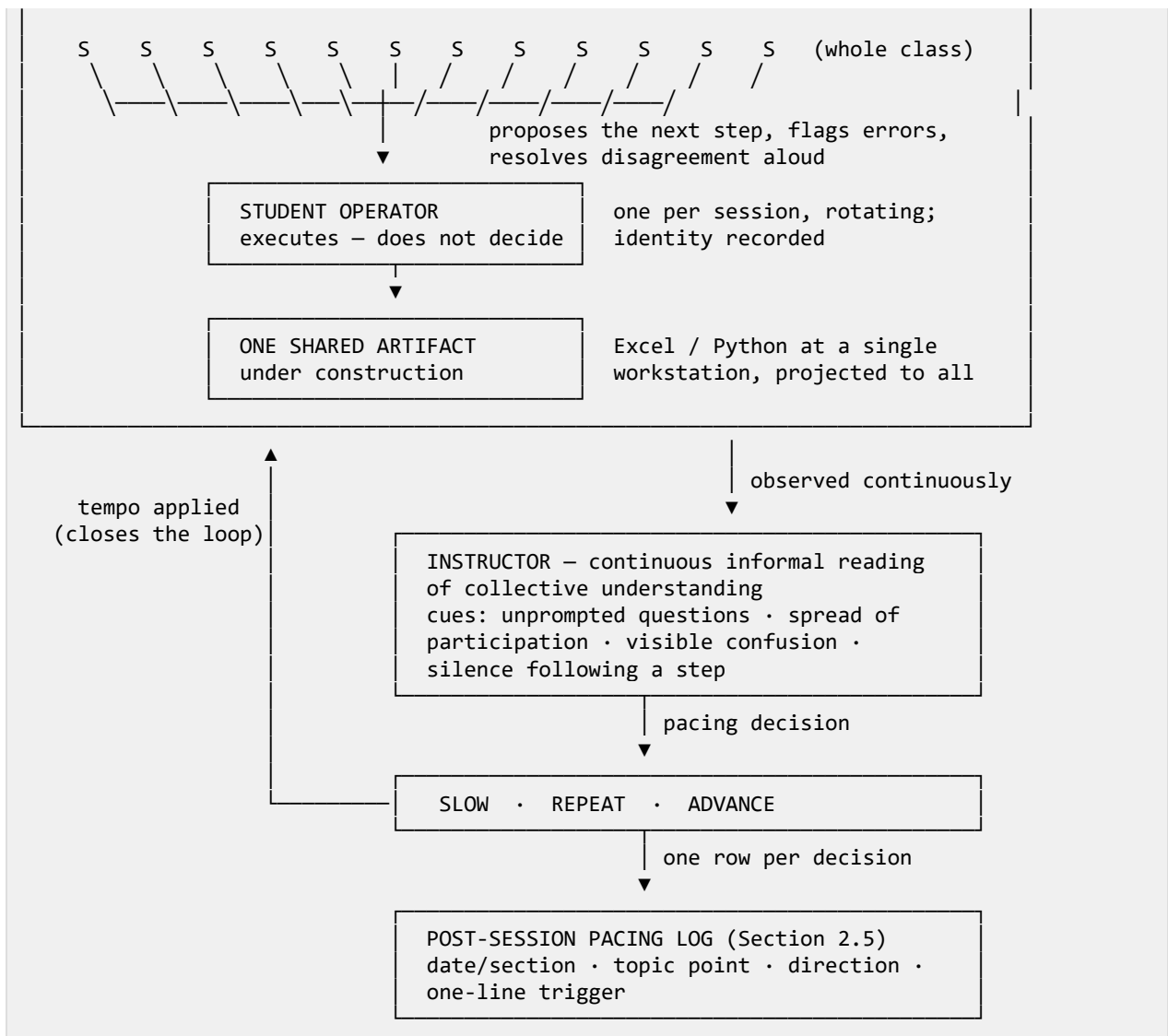


Fig. 2. The comprehension-paced collaborative problem-solving format within a single treatment session. The four constitutive properties appear as: the whole class acting as one unit (Phase 2, top); a single shared artifact under construction; the instructor's continuous rather than checkpoint-sampled reading of understanding; and pacing authority residing in that reading, applied back into the session as the closed loop on the left. The pacing log records each decision made around that loop and is the fidelity evidence reported in Section 2.5. [ASC//placeholder — to be redrawn as a publication figure.]

Table 1 sets out this contrast property by property against all four adjacent literatures, as argued in Section 1; it is not repeated here.

One consequence of defining the treatment structurally deserves explicit statement, because it is otherwise a fair objection. Since pacing is read informally rather than through a formal checkpoint instrument (Section 2.5), the treatment cannot be distinguished from attentive conventional teaching by its pacing mechanism alone; the distinction rests entirely on the structural properties above — directly observable, and either present or absent in a given session — which makes the construct *more* sharply separated from Peer Instruction and POGIL, both defined by checkpoint or role structure, rather than less.

2.3. Intervention protocol

Each treatment session followed a fixed three-phase sequence. The instructor firstly presented the underlying theory, the governing formula, and the corresponding code structure on the whiteboard, without software running. A designated student operator subsequently took the workstation and opened the tool for that session's task (Excel or Python, per the session's topic pairing). The class as a whole subsequently solved the assigned problem by directing the operator: students proposed the next step, identified errors, and resolved disagreements aloud, with the operator executing rather than deciding. Meanwhile, the instructor observed, intervened solely where the class was stuck or heading toward an unrecoverable error, and set the tempo — slowing, repeating, or advancing — according to a continuous informal reading of collective understanding drawn from verbal and visual cues (unprompted questions, the pattern and distribution of participation, visible confusion, and silence following a step).

The operator role was rotated, a different student operating in each session, rather than being filled by a fixed representative or a small volunteer pool. This choice trades one confound for another and is reported as such: rotation prevents any single student's fluency or personality from defining the treatment across the term, at the cost of session-to-session variability in operator fluency, which is left uncontrolled in the primary analysis. Operator identity was recorded per session so that this variability can be examined as a robustness check. No hypothesis is attached to it.

2.4. Control condition

Control sections received the instructor's ordinary instructor-paced lecture treatment of the same six sessions, i.e., the same syllabus, topic sequence, Excel and Python content, instructor, and summative assessments, with tempo set by the planned lesson schedule rather than by a reading of collective comprehension, and without the shared-operator arrangement. The control condition is therefore business-as-usual instruction rather than a constructed comparison treatment, which bounds the supported inference: the contrast is against ordinary practice in this course, not against another active-learning format.

2.5. Fidelity check

Since the pacing decision in the treatment condition was made on an informal reading rather than through an instrument, the consistency of treatment application cannot be established by protocol adherence alone; hence, its own evidence is strictly needed. Immediately after each treatment session, the instructor recorded a structured post-session pacing log with one row per pacing decision, capturing four fields: date and section; the problem or topic point at which the decision was made; whether the pace was slowed, repeated, or advanced; and a one-line statement of the trigger (visible confusion, an unprompted question pattern, silence, or rapid convergence). Logs were kept in a single shared spreadsheet across the three treatment sections. Descriptive statistics from the resulted log, i.e., the mean and standard deviation of pacing adjustments per session and their distribution across the three sections, are reported in Section 3 as a process measure evidencing that the treatment was applied comparably across sections rather than drifting.

Note that this is a deliberate, lower-burden adaptation of the closer journal precedent. Xu (2026), working in an analogous integrated-teaching-model study ($N = 98$), employed two independent trained observers rating six sessions on a unified five-point scale spanning participation, discussion enthusiasm, critical-thinking depth, and interactive performance, achieving inter-rater reliability above 0.85. A single-instructor, single-log design cannot claim inter-rater reliability, and the substitution is made here for feasibility within a single term with one instructor. It is consequently reported as a design trade-off and revisited in the Limitations subsection, rather than presented as equivalent.

2.6. Instruments

Four instruments were administered, three of them multi-item scales for which Cronbach's α is reported for this sample in Section 3 regardless of the α reported by the original developers. Table 2 summarizes the four. The subsections below provide the reasoning behind each selection or construction decision, which is where the present instrument choices depart from routine practice.

Table 2. Instruments, constructs, and administration schedule. Reported α is the value from the original development sample; α for the present sample is reported in Section 3 for every multi-item scale, per journal norm, and is not assumed to carry over.

Instrument	Construct (hypothesis)	Items	Response format	Source	Administration	Reported α	α , this sample
Numerical methods concept test	Conceptual performance (H1; topic-level H4)	15	4-option multiple choice; 3 items add an "I have not learned this yet" option	Purpose-built on the six-concept Delphi framework of Kaw et al. (2016)	Pretest and posttest	0.47 for the source framework's own draft instrument	Section 3
Task-specific self-efficacy scale	Performance-confidence and comprehension-confidence (H2)	12 (6 + 6 paired)	Continuous 0–100	Purpose-built, following the format of Carberry et al. (2010); factor split	Pretest and posttest	0.940–0.967 for the source instrument's subscales	Section 3, with EFA

Instrument	Construct (hypothesis)	Items	Response format	Source	Administration	Reported α	α , this sample
				per Lee et al. (2020)			
MSLQ learning-strategy subscales	Self-regulated learning as moderator (H3)	19 (12 + 4 + 3)	7-point Likert, 1–7; 4 items reverse-scored	Pintrich et al. (1991, 1993), verbatim and unmodified	Baseline only	0.79 / 0.69 / 0.76 by subscale	Section 3
Background questionnaire	Baseline equivalence; gender subgroup check on H2	4	1 open numeric, 1 categorical, 2 four-point ordinal	Purpose-built	Pretest only	Not applicable (not a scale)	Not applicable

2.6.1. Numerical methods concept test (H1)

Course performance was measured with a 15-item multiple-choice concept test, administered identically as pretest and posttest. No validated concept inventory for numerical methods was available for direct adoption: the most developed instrument in the field reached a third-draft 24-item form with classical item statistics, albeit with an overall reliability of $\alpha = 0.4725$, below the conventionally acceptable range, and was explicitly described by its developers as not yet finalized (Kaw et al., 2016). Note that this absence is a recognized field-wide gap in quantitative-computational teaching rather than a local one, having been reported in adjacent domains on the same grounds (Guzman et al., 2019). The instrument employed here was therefore purpose-built, yet scaffolded onto the expert-validated content structure of that work rather than onto instructor judgment. Items were written against the six core concepts converged upon through Kaw et al.'s four-round Delphi process with 13 expert instructors, i.e., the relationship of the Taylor series to numerical methods, transformation between representations of a method, monitoring and interpreting convergence, converting a mathematical model into algorithm-ready form, identifying all solutions or their absence, and selecting an appropriate technique, respectively, with each concept addressed by at least two items and each course session contributing two or three items. Table 3 provides the full mapping of sessions to topics, tools, target concepts, and items for both pre/post instruments.

Table 3. Course sessions mapped to topics, computational tools, target concepts, and instrument items. Concept labels A–F follow the six-concept framework of Kaw et al. (2016): A, Taylor series basis; B, transformation between representations; C, monitoring and interpreting convergence; D, converting a model into algorithm-ready form; E, identifying all solutions or their absence; F, selecting an appropriate technique.

Session	Topic	Tools	Concepts	Concept-test items	Self-efficacy items
1	Linear equation systems: Gauss–Seidel, Jacobi	Excel formula / <code>np.linalg.solve</code>	C, B	1–2	1–2
2	Linear systems for 1D steady-state conduction	as Session 1, applied case	D	3–4	3–4
3	Nonlinear equations, analytic	Horner's method / <code>np.roots</code>	E, F	5–6	5–6
4	Nonlinear equations, numerical	Excel bisection/secant/Newton; <code>scipy.optimize.fsolve</code>	A, C, B/F	7–9	7–8
5	Nonlinear systems: extent of reaction	<code>scipy.optimize.fsolve</code> / root	D, C, E	10–12	9–10
6	Regression: kinetic parameter estimation	Excel Solver; <code>scipy.optimize.least_squares</code>	D, F, C/E	13–15	11–12

Concept-test items 9, 12, and 15 are the three black-box items described below; self-efficacy items are paired within each session, the odd-numbered item measuring performance-confidence and the even-numbered item comprehension-confidence.

Three items, one each in the final three sessions, probe whether conceptual understanding survives when the algorithm is concealed behind a library call, which is the sharpest available test of the compound-load argument

advanced in Section 1. These three items offer an explicit "I have not learned this yet" option in place of forcing a guess among four substantive alternatives, since students possess no prior exposure to the tools concerned at pretest; the option is scored as incorrect, whereas its selection rate is tallied separately as a baseline-novice descriptive. Total score (0–15) serves as the covariate and outcome pair for H1, and the six session-tagged item clusters serve as the topic-level sub-scores for H4 without requiring separate tagging of course examinations. Near-floor pretest scores are anticipated rather than treated as a defect, since the content is genuinely new to the students. Consequently, the pretest covariate may exhibit restricted variance and function principally as a test of baseline equivalence across sections rather than as a substantive adjustment.

2.6.2. Task-specific self-efficacy scale (H2)

Self-efficacy was measured with a 12-item purpose-built scale administered as pretest and posttest. Two design decisions depart from the common practice of adopting a generic course self-efficacy subscale, and both follow directly from the measurement literature. Firstly, the scale is task-specific rather than course-general and employs a continuous 0–100 response format anchored at "not at all confident," "moderately confident," and "completely confident," following the engineering design self-efficacy instrument of Carberry et al. (2010), whose subscales reached $\alpha = 0.940$ – 0.967 and which discriminated self-efficacy across high, intermediate, and low engineering-experience groups; that continuous format was itself adopted on evidence that continuous scales predict performance better than ordinal Likert alternatives. Secondly, and more consequentially here, the scale deliberately separates confidence in *producing a correct result* from confidence in *understanding why the method works*. In an active, blended, and collaborative engineering course ($N = 681$), the standard single-factor self-efficacy structure did not fit ($RMSEA = 0.111$, $CFI = 0.811$), splitting instead into performance-expectation and comprehension factors, of which solely the performance factor predicted course grade (Lee et al., 2020). Since the argument motivating the present intervention is fundamentally about comprehension rather than examination confidence, an instrument collapsing the two would risk an apparent dissociation between H1 and H2 which is a measurement artifact rather than a finding.

The scale therefore mirrors the concept test's six-session structure, contributing one performance-confidence item and one paired comprehension-confidence item per session, for instance confidence in obtaining the correct temperature profile for a discretized one-dimensional conduction problem, paired with confidence in understanding how discretizing that physical problem yields the linear system being solved. Performance-confidence and comprehension-confidence scores are computed as separate subscale means and reported separately unless an exploratory factor analysis on the present sample supports collapsing them. That analysis is reported in Section 3 as part of the instrument's validation for this sample, and its outcome is treated as a substantive result bearing on Lee et al.'s finding rather than as a preliminary step.

2.6.3. Self-regulated learning baseline (H3)

Self-regulated learning was measured once, at baseline, using three learning-strategy subscales of the Motivated Strategies for Learning Questionnaire, reproduced verbatim from the original manual (Pintrich et al., 1991, 1993): Metacognitive Self-Regulation (12 items, $\alpha = 0.79$ in the original sample), Effort Regulation (4 items, $\alpha = 0.69$), and Peer Learning (3 items, $\alpha = 0.76$), for 19 items in total, each rated on the original seven-point scale from "not at all true of me" to "very true of me," with four negatively worded items reverse-scored. Items were used unmodified rather than adapted to this course's tools, so that the original reliability evidence carries over and no re-validation burden is incurred that a sample of this size could not discharge.

Two selection decisions warrant justification. Firstly, these three subscales were chosen because they correspond to the components of self-regulation which a comprehension-paced whole-class format plausibly substitutes for or recruits, i.e., monitoring one's own understanding, persisting through difficulty, and learning through classmates, respectively, rather than because they are the instrument's most commonly reported subscales. Indeed, the same three components behave as separable constructs longitudinally (Esnaashari et al., 2025) and, in the case of the first two, discriminated substantially between instructional conditions in an undergraduate engineering laboratory ($d = 0.655$ and $d = 0.826$, respectively; Cepeda-Argüelles et al., 2026). Secondly, the Online Self-Regulated Learning Questionnaire was considered and rejected on two grounds. Its canonical six-factor structure has not replicated consistently: a cross-cultural re-validation found a seven-factor model fitting significantly better than the canonical six ($N = 571$; scaled $\chi^2(6) = 208.91$, $p < 0.001$) and reported that lower-SRL students systematically overestimate their own regulation on self-report (Na et al., 2025), whereas the largest published exploratory analysis to date ($N = 1,434$) failed to recover the six-factor structure at all, retaining merely two factors, against a documented range of subscale reliabilities across international validations running from $\alpha = 0.41$ to $\alpha = 0.96$ (Rufini et al., 2021). Note that the overestimation finding is a specific threat to the present use of self-regulation as a moderator, since it would bias the moderator toward the null in exactly the lower-SRL subgroup which H3 concerns. Independently of its psychometrics, that instrument's items are worded around asynchronous online study behavior, which maps poorly onto a synchronous in-person classroom.

Administering this scale merely once, at baseline, treats self-regulated learning as a moderating individual difference rather than as an outcome. This is a design choice with a known cost: since the engineering-laboratory evidence cited above found metacognitive self-regulation and effort regulation differing between instructional conditions, a single baseline reading cannot detect whether the intervention itself shifts these components, and no such claim is made.

2.6.4. Background questionnaire

A four-item background questionnaire was administered once, within the pretest packet: cumulative grade point average, gender, prior experience utilizing spreadsheets for engineering calculation, and prior experience with Python or another programming language, the latter two on four-point ordinal scales from none to extensive. Grade point average and the two prior-exposure items support the baseline-equivalence test across non-randomized sections described in Section 2.1 and enter the analysis as covariates. Gender is retained for a subgroup check on the self-efficacy outcome, motivated by evidence that women in engineering populations report self-efficacy-lowering formative events significantly more often than men, in a context where problem-solving is publicly visible (Chen et al., 2023).

2.6.5. Administration

The pretest packet, comprising the concept test (15 items), the self-efficacy scale (12 items), the self-regulated learning subscales (19 items), and the background questionnaire (4 items) for 50 items in total, was administered to all five sections prior to the first of the six sessions. The posttest packet, comprising the concept test and the self-efficacy scale identical in wording to their pretest form for 27 items in total, was administered to all five sections after the sixth session. [DATA NEEDED — administration mode (online form versus paper) and measured completion time are not yet fixed; a timing check should precede the pretest, since a 50-item packet may exceed the intended session allowance.] Content alignment of the two purpose-built instruments was reviewed by a colleague against the item-to-concept mapping prior to first administration. A single such review, rather than a full pilot with item analysis, is what the term's timing permitted, and is reported as a limitation.

2.7. Analysis plan

The primary analyses for H1 and H2 are analyses of covariance in which condition is the factor, the corresponding pretest score is the covariate, and the posttest score is the outcome. Since condition was assigned by section rather than by student, section is modelled as a random effect nested within condition; thus, the primary tests are mixed-effects models rather than student-level ANCOVA, and section-level clustering is not treated as independent variance. Baseline equivalence across the five sections is examined firstly, utilizing the pretest concept-test score, pretest self-efficacy scores, grade point average, and the two prior-exposure items. Covariates exhibiting section-level imbalance are retained in the primary models.

Three secondary analyses are reported as exploratory, with that status stated rather than implied. H3 adds a self-regulated learning $\times$ condition interaction term to the primary model, run separately for each of the three subscales, and is underpowered by construction, since an interaction test on $N \approx 200$ split three sections to two is unlikely to detect any but a large moderation effect. Note that the directional expectation is compensatory rather than amplifying, i.e., that external pacing structure benefits lower-self-regulation students disproportionately, following the pattern reported for an adaptive structuring intervention in online learning (Pogorskiy and Beckmann, 2023). H4 repeats the primary analysis on the six session-tagged concept-test sub-scores, with reduced per-topic item counts and correspondingly wider intervals. H5, a time $\times$ condition mixed-effects model on the self-efficacy trajectory, is reported solely if a mid-term measurement point proves feasible, and is omitted otherwise.

Effect sizes (Cohen's d for pairwise contrasts, partial η^2 for model terms) are reported alongside significance tests for every primary and secondary analysis, with confidence intervals. Following the observation that a course-level intervention may move pass rates more than continuous achievement scores (Walker and Warfa, 2017), the H1 outcome is additionally reported dichotomously where the course's own passing threshold applies. Moreover, Cronbach's α is reported for this sample for each multi-item scale and, for the concept test, item difficulty indices, point-biserial correlations, and alpha-if-item-deleted are reported following the classical-test-theory approach employed in the instrument's source framework (Kaw et al., 2016). Table 4 states the model, effect-size statistic, and inferential status committed to for each hypothesis in advance of analysis.

Table 4. Hypotheses, outcome measures, models, and inferential status. Primary and exploratory status is fixed here rather than assigned after results are seen.

	Outcome	Model	Effect size	Status
H1	Concept-test posttest score (0–15); additionally dichotomized at the course passing threshold	Mixed-effects ANCOVA; condition fixed, section nested within condition random; pretest covariate	Cohen's d , partial η^2 , with CI	Primary
H2	Performance-confidence and comprehension-confidence subscale means (0–100), modelled separately	As H1, run on each subscale	Cohen's d , partial η^2 , with CI	Primary

	Outcome	Model	Effect size	Status
H3	H1 and H2 outcomes	H1/H2 model plus an SRL × condition interaction, run separately for each of the three MSLQ subscales	Partial η^2 for the interaction, with CI	Exploratory; underpowered by construction
H4	Six session-tagged concept-test sub-scores	H1 model repeated per session cluster	Cohen's d per cluster, with CI	Exploratory; reduced per-topic item count
H5	Self-efficacy trajectory	Time × condition mixed-effects model with repeated measures	Partial η^2 for the interaction	Optional; reported only if a mid-term measurement point proves feasible, omitted otherwise

2.8. Ethics and data handling

[DATA NEEDED — institutional review board approval status is not yet confirmed; approval reference and date to be inserted prior to submission.] Participation in the surveys was voluntary and independent of course grading, and informed consent was obtained. Both conditions received the full syllabus, the same instructional content, and the same assessments; hence, no section was disadvantaged in course outcomes by its condition assignment. Survey responses were linked to course-performance records through a study identifier rather than student identity, identifying information was removed prior to analysis, and data were stored and processed in accordance with institutional data-protection requirements.

Please review this section for unintended similarity to source texts before proceeding.

Open flags carried into this draft

- The prior [CITATION NEEDED – VERIFY AUTHOR NAMES] flag on the self-paced-online-procrastination claim (§4) is now resolved: that sentence is replaced with full-text-read sources (Lim, 2016; Chiu et al., 2024; Pogorskiy and Beckmann, 2023) carrying the same argument with confirmed citations. The "flipped-classroom ACS/RSC review" source remains unneeded in this draft (Lewin and Barzilai, 2023 carries the flipped-classroom point on its own, full-text-read and directly on-topic).
- All citations in this section now trace to [LIT: . . .] tags with full-text sourcing confirmed in the Scratchbook Reference List — no open citation flags remain in the Introduction.
- **Materials and methods (added 2026-09-04 16:37):** two [DATA NEEDED] flags remain inline — (i) §2.6.5, pretest/posttest administration mode and measured completion time (the packet is 50 items, above the ~25–35 min target originally assumed); (ii) §2.8, IRB approval reference and date. Both are researcher actions, not drafting gaps. Two further statements are written as planned protocol and must be verified against actual implementation before submission: that operator identity is in fact recorded per session (§2.3), and that the colleague content-alignment review of the concept test and self-efficacy scale actually took place before first administration (§2.6.5). Koyande and Wong (2026) is cited nowhere in this section — the section-confound-check precedent it supplies is realized in §2.7's baseline-equivalence step rather than cited, since the paper has not been re-read at Methods-drafting depth ([UNVERIFIED] in the Scratchbook).
- No citation appears in the Methods that lacks a full-text-confirmed Scratchbook Reference List entry. The Methods section carries no [CITATION NEEDED] flags.
- **Display items (added 2026-09-04):** Figs. 1 and 2 are ASCII placeholders and must be redrawn as publication figures before submission; Fig. 1 additionally needs participant-flow and attrition numbers once data collection completes. Table 2's "α, this sample" column and Table 4's results are unfillable until data exist — both are written to be completed rather than revised. If the journal's figure/table allowance turns out tighter than the 2–7 / 1–9 range observed in the sample, Tables 3 and 4 are the two intended for demotion to supplementary material; Table 1 and Fig. 2 carry novelty claims and should stay in the main text.

Change log

- 2026-09-05 18:23 (§1 citation addition) — Introduction §1's opening sentence ("Computational competence is strictly needed...") previously carried no citation; it now cites Udugama et al. (2022) (three linked institutional surveys showing near-unanimous alumni/expert/staff support for digitalisation in the ChemE curriculum, with digital-model competency expected above statistical/ ML content), Johnson and Singh (2018) (illustrative curriculum practice already embedding simulation software into core technical content), and Udugama et al. (2023) (IChemE-institution survey rating Excel and scripting languages, not compiled languages or immersive platforms, as best-suited for teaching core quantitative subjects because of low mathematical/programming overhead). All three folded into the existing sentence via appositive/parenthetical clauses rather than as new sentences, keeping §1 at its existing 8-sentence ceiling. The Udugama et al. (2023) finding was placed to lend independent field-consensus support to the paragraph's own subsequent compound-cognitive-load argument (low-overhead tool choice keeps the tool from becoming the dominant cognitive demand), rather than merely

- restating the opening claim. references.md and literature/library.md updated accordingly (34 entries; both new sources marked "manuscript Intro (Draft 1)").
- 2026-09-04 (formatting) — Reformatted every manuscript-body paragraph (Introduction and Materials and methods prose, figure/table captions, and the draft-status blockquote) to a single unbroken line in the markdown source, per researcher preference — no content, wording, or citation changed. Tables, ASCII figure placeholders, and headings are unaffected. Logged as a proposed ARTi-writing feature in `_ARTi\skill-improvement-ideas.md`.
 - 2026-09-04 17:03 (deep gtmk-voice pass + concision) — Audit found the prior draft carried only two gtmk mechanics (the we-placement rule and figure/table pointer sentences); the distinctive machinery was absent. Applied in this pass across the whole Introduction and Methods: **(i) connective openers** on nearly every non-initial sentence (However, Moreover, Note that, Indeed, Nevertheless, Similarly, Consequently, Meanwhile, Differently, Hence); **(ii) paragraph rhythm** — long subordinated declaratives closed by a short verdict sentence ("Neither feature renders the design randomized.", "No hypothesis is attached to it.", "Fully asynchronous online formats remove instructor supervision altogether."), with every paragraph brought inside the 4–8 sentence band (Intro ¶4 was ~15 sentences, now 8); **(iii) lexicon** — subsequently for then, prior to for before throughout Methods, employ/utilize for instruments and methods, i.e. series (with the respectively list-closing tic held to three uses, since seven overshoot the skill's ~1-in-8 imperfection budget), so-called for scare quotes, ; thus, / ; hence, causal chains, owing to, albeit, whereas; **(iv) intensity ladder** — unmodified changes given a modifier (substantially beyond, notably more, discriminate substantially); **(v) commitment gradient** — hedges stripped from Methods procedure and numeric readouts, kept on mechanism claims (plausibly supplies, may crowd out, may exhibit); **(vi) the non-native fingerprint**, previously at zero occurrences: the resulted evidence (Intro ¶6) and the resulted log (§2.5) — two of the prescribed 2–3 per manuscript, leaving one for Results — plus strictly needed (¶1, §2.5), possess their own handicaps (¶2), similar with (¶5), is of practical advantage (¶6), albeit mid-sentence, and which-for-that throughout, at roughly one visible imperfection per eight sentences. **Results tense is hereby fixed as present** (the skill's default, from the corpus's most recent paper) and must be held for Sections 3 onward. Elsevier author–year citations retained per Journal Profile Block B delta. Redundancy removed in the same pass, with no loss of substance or citation: the Teixeira et al. (2018) description appeared near-verbatim in both Intro ¶3 and ¶5 and is now stated once, in ¶5 where the closest-neighbor argument lives; §2.2's paragraph re-describing Peer Instruction, POGIL, flipped, and asynchronous formats duplicated Intro ¶3–¶5 and Table 1 entirely and is replaced by a one-sentence cross-reference; §2.2's enumeration of the four constitutive properties was restated a third time in the Fig. 2 caption and is now compressed to a single clause pointing at Table 1; the Crouch and Mazur physics-implementation clause in ¶3 was dropped as subsumed by the Balta et al. (2017) meta-analysis; and the Cepeda-Argüelles effect sizes ($d = 0.655$, $d = 0.826$) were moved out of the Introduction into §2.6.3, where they actually justify the subscale selection. Net effect: roughly 450 words of prose removed (order 10 % of the Introduction and Methods narrative, which now stands at 6,418 words including display items), with every citation, hypothesis, display item, and open flag preserved. No [DATA NEEDED] flag was resolved or silently dropped in this pass.
 - 2026-09-04 — Draft 1 of the Introduction created from the Scratchbook Introduction section, following the Manuscript Blueprint's six-paragraph structure and the generalized gtmk-voice mechanics (Elsevier author–year citation style per Journal Profile Block B delta, not the voice skill's default bracketed-numeral style).
 - 2026-09-04 (revision) — Revised all six paragraphs using the 17 full-text papers fed into the Scratchbook's second Literature Feed Protocol pass. ¶1 gained Robins (2019)'s formal five-domain/cognitive-load grounding for the compound-load claim. ¶2 gained Chen et al. (2023) as domain-specific Bandura support, with an explicit caution that visible performance moments can lower as well as raise self-efficacy. ¶3 gained meta-analytic effect-size evidence for both Peer Instruction (Balta et al., 2017; Aldalou et al., 2026) and POGIL (Walker and Warfa, 2017; Udu et al., 2020), plus the same-discipline near-miss Teixeira et al. (2018). ¶4's procrastination citation-needed flag is now resolved with Lim (2016), Chiu et al. (2024), and Pogorskiy and Beckmann (2023), which sharpen the self-regulation argument feeding into H3. ¶5 gained Teixeira et al. (2018) as the closest-in-discipline (not closest-in-venue) contrast paper, alongside the existing four closest-neighbor citations. ¶6 (objective/contribution) unchanged.
 - 2026-09-04 16:37 (MSLQ enrichment) — Introduction ¶4 revised to name and decompose the self-regulated-learning construct it previously argued about only implicitly: the MSLQ's separation of motivational beliefs from learning strategies, and of metacognitive self-regulation / effort regulation / peer learning as distinct subscales (Pintrich et al., 1991, 1993), plus evidence that these components move differentially over a term (Esnaashari et al., 2025) and that the first two discriminated between instructional conditions in an undergraduate engineering laboratory with $d = 0.655$ and 0.826 (Cepeda-Argüelles et al., 2026). This closes the Scratchbook's standing note that no canonical SRL-theory citation had been wired into the Introduction, and it now motivates H3's three-subscale choice in the Introduction rather than only in the Methods. Antwi et al. (2026) and Callejo-Martin et al. (2026) were deliberately *not* added to the Introduction — both are instrument-wording/psychometric sources whose contribution to this project was recovering verbatim MSLQ item text, which is a Methods matter and is superseded there by the 1991 primary source. No other paragraph changed.
 - 2026-09-04 16:37 (Methods) — Materials and methods drafted in full as §2, following the Manuscript Blueprint's eight-subsection shape (participants → operational definition → protocol → control → fidelity → instruments → analysis → ethics), with the Scratchbook's "Distinguishing the Treatment from Just Good Teaching" material folded into §2.2's closing paragraph and instrument detail drawn from the four instrument files finalized earlier the same day. Written in past tense per the final manuscript's form, with a draft-status note flagging that the study is not yet executed. Block C norms addressed explicitly: per-sample Cronbach's α promised for every multi-item scale (§2.6, §2.7), effect sizes with confidence intervals for all primary and

- secondary analyses (§2.7), the Xu (2026) two-observer fidelity precedent named as the standard this study deliberately simplifies rather than silently omits (§2.5), and an explicit ethics/data-handling subsection (§2.8).
- 2026-09-04 (display items) — Two figures and four tables added, all cited in body text at first mention and numbered by order of first citation. **Fig. 1** (§2.1) study design, condition assignment, and measurement schedule; **Fig. 2** (§2.2) the single-session classroom mechanism with the pacing feedback loop — the schematic Block C flags as a journal norm for papers proposing a named pedagogical model (Xu's "Four-in-One" figure; Zhang's experiential-learning-cycle figure). **Table 1** placed in the Introduction rather than Methods, immediately after the gap statement (§5), since it is the artifact that makes the C3 novelty claim checkable — property-by-property contrast against all four adjacent literatures, cross-referenced from §2.2. **Table 2** (§2.6) instrument summary, with an α column deliberately left pointing to Section 3 rather than filled from source studies. **Table 3** (§2.6.1) session→topic→tool→concept→item mapping, the content- validity evidence for the concept test and the definition of H4's clusters. **Table 4** (§2.7) hypothesis→model→effect-size→status map, which pre-commits the primary/exploratory split before results exist. Both figures are ASCII placeholders for redrawing; Fig. 1's caption notes that participant flow and attrition are to be added CONSORT-style once data collection completes. Display-item count (2 figures, 4 tables) sits inside the journal sample's observed range (2–7 figures, 1–9 tables) and leaves figure budget for Results.